INFORMATION TRUST EXCHANGE GOVERNING ASSOCIATION

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NEWSHARE NETWORK

Technical Architecture and Specification

A Federated Identity, Single Sign-On and Information Commerce Network

Governing Authority: Information Trust Exchange Governing Association (ITEGA)
Network Operator / Technical Lead: Clickshare Service Corp.
Pilot Host Institution: Donald W. Reynolds Journalism Institute, University of Missouri

Version 1.0 Draft — February 2026

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DRAFT — For discussion with technical partners and funders

Executive Summary

The Newshare Network is a proposed federated identity, single sign-on, and information commerce network designed to serve the news publishing industry. It is governed by the Information Trust Exchange Governing Association (ITEGA), a U.S. 501(c)3 nonprofit corporation incorporated in California in 2017. Its architecture draws on a working reference implementation built by Clickshare Service Corp. in 1996 and patented in 2008 (U.S. Patent No. 7,324,972, now expired). The underlying ideas are in the public domain.

The network enables a reader to create one account at a participating “home base” publisher or service provider, and to be recognized — without re-registering — at any other participating publisher on the network. Publishers share users without sharing identity data. The system logs all value-exchange events (content access, advertising views, subscription credits) through a neutral authentication and logging service, and settles payments periodically across the banking ACH network.

> *The Newshare Network is what the internet’s information economy needed in 1996 and still lacks in 2026: a nonprofit-governed, privacy-by-design layer for identity, access, and fair payment across independent publishers — with no platform intermediary taking the margin.*

The architecture has four parties: (1) the End User, (2) the End User’s Home Base (Identity Service Provider), (3) the Content Provider (Publisher), and (4) ITEGA-licensed Authentication, Logging and Settlement Services. ITEGA itself governs the rules; it does not operate the services.

The immediate goal is a proof-of-concept pilot involving a small consortium of independent Missouri newspapers, hosted at the Reynolds Journalism Institute (RJI) at the University of Missouri. The pilot would demonstrate the end-to-end architecture: a user authenticates once at a home base, visits a second participating publisher, is recognized without re-registering, and a transaction record is logged and settled.

Key Design Principles

- Privacy-by-design: The authentication and logging service never knows a user’s real identity — only an opaque alphanumeric identifier assigned by the home base.
- No central identity database: User data is managed locally at the home base. The network uses distributed, DNS-like architecture.
- No cookies required: Authentication state is passed via token in HTTP headers, not browser cookies — a design principle established in the 1996 Clickshare reference implementation.
- Wholesale-retail pricing: Publishers set wholesale content prices; home-base service providers set retail prices for their users, capturing margin as a business model.
- Pairwise pseudonymous identifiers: Each user receives a different anonymous network ID for each publisher they visit, preventing cross-site correlation.
- Governed, not owned: ITEGA sets the rules and certifies participants. Commercial operators run the services. Profit goes to operators and publishers, not to ITEGA.

1. Background and Context

1.1 The Problem

The internet's information economy has consolidated around a small number of platform intermediaries — principally Google and Meta — who have captured the majority of digital advertising revenue that once supported independent journalism. Publishers have lost control of their user relationships. Readers have lost control of their personal data. Advertisers pay for opaque, fraud-prone inventory. Democracy suffers as local news withers.

The technical root cause is the absence of a shared, neutral infrastructure for identity, access and payment across independent publishers. Each publisher operates a silo. Readers must register separately at each site. No mechanism exists for a reader's home publisher to vouch for them at another publisher, or for content access to be logged and settled across the network without a platform intermediary.

1.2 The History

In 1993, Bill Densmore founded what became Clickshare Service Corp. to solve precisely this problem. In 1996, Clickshare stood up a working reference implementation of a four-party federated identity and micropayment network — the Token Validation Service (TVS) — built on a modified NCSA httpd web server. The system enabled single sign-on, per-click micropayments, audience measurement, and periodic ACH settlement across independent publishers, without cookies and without a central identity database.

U.S. Patent No. 7,324,972 ("Managing Transactions on a Network: Four or More Parties") was granted in 2008 and expired in 2022. Its architectural concepts are now in the public domain. The patent's co-inventors are Bill Densmore and Richard Lerner, who remain the principal architects of the Newshare Network.

ITEGA was incorporated as a California 501(c)3 nonprofit in 2017 to serve as the governing authority for a modernized version of the Clickshare architecture. Bill Densmore was an academic-year fellow at the Donald W. Reynolds Journalism Institute at the University of Missouri in 2008–2009, where the ITEGA concept was further developed. RJI has supported the project over the years and is the proposed institutional host for the proof-of-concept pilot.

1.3 Why Now

Three developments make 2026 the right moment to build this:

1. The regulatory environment has shifted decisively toward user privacy. GDPR, CCPA, and the collapse of third-party cookies have created both the mandate and the market incentive for a privacy-by-design identity infrastructure.
2. The standards layer now exists. OpenID Connect (OIDC), OAuth 2.0, JSON Web Tokens (JWT), and W3C Verifiable Credentials provide a modern protocol foundation that maps almost directly onto the original TVS architecture. What once required custom C-language wire protocols can now be composed from open-source libraries.
3. The news industry is in genuine crisis and increasingly open to collaborative solutions. The failure of advertising-only models is no longer deniable. Publishers need alternative revenue streams, and readers are willing to pay for trusted local journalism — if the friction of registration and payment is reduced.

2. System Architecture

2.1 The Four-Party Model

The Newshare Network involves four distinct parties, each with defined roles and relationships:

Party	Role and Description
1. End User	A reader or consumer who registers once at their chosen home base. The user is identified within the network only by an opaque alphanumeric identifier. The user never has a direct relationship with ITEGA.
2. Home Base (Identity Service Provider — IdSP)	A publisher, news organization, ISP, library, or other trusted entity where the user holds their primary account. The home base authenticates the user, manages their profile and privacy preferences, and vouches for them across the network. The home base is the only party that knows the user's real identity. It bills the user and pays publishers for content accessed.
3. Content Provider (Publisher / CP)	A news organization or information service that publishes content on the network. Publishers accept authenticated users from any home base without requiring those users to register. They set wholesale prices for their content and receive payment through the settlement service.
4. Auth / Logging / Settlement Service (ALS)	A neutral, ITEGA-licensed infrastructure service that: (a) validates authentication tokens in real time, (b) logs all value-exchange events against opaque user IDs, and (c) periodically settles payments among home bases and publishers via ACH. The ALS never has access to personally identifiable information.

> *ITEGA itself is none of these four parties. ITEGA is the governing authority that sets rules, certifies participants, and enforces standards. It licenses operators to run the ALS. Profit flows to operators and publishers, not to ITEGA.*

2.2 The Authentication Flow

The following describes the authentication sequence when a user visits a publisher where they are not directly registered. This maps to a standard OpenID Connect Authorization Code flow with home-site discovery added.

Step	Description
1	User clicks a content link at Publisher B (a network member where they have no account).
2	Publisher B's server detects no local session and redirects to its authentication service, which presents a "Network Login" option alongside local login.
3	User selects Network Login. The authentication service checks for a previously set home-site cookie from the ITEGA network server. If found, home-site discovery is automatic. If not, the user selects their home base from a list.
4	The network server redirects the user's browser to their home base (IdSP) authentication controller.

5	The home base authenticates the user in its normal manner (username/password, passkey, or other). It then generates or retrieves the appropriate pairwise Network User ID (networkUserId) for this specific user+publisher combination.
6	The home base passes the networkUserId and a NetworkGroupId (encoding the user's subscription tier and access rights) to the network server, which redirects back to Publisher B.
7	Publisher B maps the NetworkGroupId to its local access controls and serves the content. It records the transaction to the ALS logging service.
8	At settlement time (daily, weekly, or monthly), the ALS aggregates all logged events, computes net debits and credits by home base and by publisher, and initiates ACH transfers.

Once authenticated, Publisher B may cache the authentication for a session period, avoiding repeated redirects. If the user's session expires, the network transparently returns them to their home base for re-authentication.

2.3 The Pairwise Pseudonymous Identifier

A critical privacy feature of the Newshare Network is the pairwise pseudonymous identifier scheme, co-designed by Richard Lerner. The home base maintains a mapping from its internal user IDs to unique networkUserIds assigned per remote publisher. If user "Susan" (internal ID 12345 at her home base) visits Publisher B, she is assigned networkUserId "NNN-987643" where NNN is Publisher B's network ID. When she visits Publisher C, she receives a completely different networkUserId.

This means Publisher B and Publisher C each see a different opaque identifier for Susan, making cross-site correlation technically impossible without the home base's cooperation. The ALS also sees only these opaque IDs and cannot correlate them to a real person. This design implements what the W3C OIDC specification calls a Pairwise Pseudonymous Identifier (PPID), now a recognized standard.

The home base can unlink a user from one or all networkUserIds at any time — effectively making the user "disappear" from a publisher's records without any action by the publisher.

2.4 The NetworkGroupId

The NetworkGroupId is a bitmask value that encodes a user's subscription status and access rights in a standardized format that all network members recognize. The home base maps from its internal subscription records to the network group schema; publishers map from the network group schema to their local access controls. This allows a wide variety of subscription models to interoperate without exposing proprietary business logic.

Bit Value	Group Name / Description
0	Anonymous — accessing without login (pre-registration meter)
1	Group Account — IP-based or access-key access
2	Registered — logged in with individual account
4	Print Subscriber
8	Digital / Web Subscriber
16	Data / Special Content Subscriber
1024	Complimentary Subscriber
2048	Controlled (free) Subscriber

4096	Paid Subscriber
8192	Trial Subscriber
16384	Site / Group Subscriber (corporate, university, library)

Group IDs can be combined bitwise. For example, a user who is both a Paid Subscriber (4096) and a Print Subscriber (4) would have NetworkGroupId 4100. Additional group IDs may be added by ITEGA governance as needed.

3. Protocol Specifications

3.1 Identity and Authentication Layer

The Newshare Network uses OpenID Connect (OIDC) 1.0 on top of OAuth 2.0 as its core identity federation protocol. OIDC is the dominant SSO standard in deployment today, with broad library support across all major programming languages and platforms. It maps directly onto the original TVS token architecture: the OIDC ID Token (a signed JWT) is the modern equivalent of the TVS authentication token.

Function	Protocol / Standard
Core SSO federation	OpenID Connect 1.0 (Authorization Code Flow)
Token format	JSON Web Token (JWT, RFC 7519)
Home-site discovery	OIDC Discovery (/.well-known/openid-configuration) + WebFinger (RFC 7033)
Pairwise user IDs	OIDC Pairwise Pseudonymous Identifiers (PPID, per OIDC Core spec section 8)
Rich identity claims	W3C Verifiable Credentials (VC Data Model 2.0) for portable, user-controlled attributes
Token encryption	JSON Web Encryption (JWE, RFC 7516) for sensitive claim protection
Subscriber tier encoding	Custom NetworkGroupId claim in OIDC ID Token (bitmask, see Section 2.4)
User profile attributes	Schema.org/Person + Internet2 eduPerson attribute schema
Content rights tagging	Really Simple Licensing (RSL) standard (rslstandard.org, 2025)
Transport security	TLS 1.3 mandatory for all network connections

3.2 User Attribute Schema

User attributes are stored at the home base and shared with the network only as permissioned by the user. The schema separates personally identifiable information (PII) from anonymizable demographic attributes. The ALS never receives PII; it receives only the opaque networkUserId and the NetworkGroupId.

NETWORK-LEVEL ATTRIBUTES (ACCOMPANY ALL AUTHENTICATED REQUESTS)

Attribute	Description / Notes
networkUserId	Globally unique, pairwise pseudonymous identifier. Format: [HomeBaseID]-[OpaqueToken]. Minimum attribute for logging.
homeBaseId	Network ID of the user's home base (IdSP). Analogous to credit card issuer ID.
networkGroupId	Bitmask encoding subscription tier and access rights (see Section 2.4).

sessionToken	Signed JWT issued by the ALS. Valid for configurable session duration.
pubMbrId	Network ID of the content-serving publisher (for logging and settlement).

PREFERENCE-LEVEL ATTRIBUTES (GOVERN ALL REQUESTS)

Attribute	Description / Notes
privacyLevel	User-set: open, limited, or private. Controls what demographic attributes are shared.
adPreference	User-set: full ads, links only, no ads.
doNotTrack	Boolean. If true, no cross-site behavioral data is logged beyond billing minimum.
parentalControl	Boolean. If true, content filtering rules apply.
ageRange	Enum: <13, 13-17, 18-24, 25-34, 35-44, 45-54, 55-64, 65+. Enables COPPA compliance without transmitting birthdate.

OPTIONAL IDENTITY ATTRIBUTES (SHARED ONLY WITH USER PERMISSION)

Attribute	Type / Format
nickname	String (display name, not necessarily real name)
email	String, 255 chars. PII. Shared only with explicit consent.
gender	Enum: preferNotToSay, male, female, genderQueer, other
postalCode	String, 15 chars
country	ISO 3166-1 alpha-2 (2-letter code)
language	BCP 47 language tag
income	Enum ranges: <20K, 20-34K, 35-49K, 50-74K, 75-99K, 100-149K, 150-199K, 200K+
education	Enum: primary, secondary, postSecondary
employment	Enum: notEmployed, looking, partTime, fullTime, retired, disabled
interests	Array of freeform keywords/phrases (user-supplied topical interests)

3.3 Content Tagging for Rights and Pricing

Publishers tag content with metadata that the network uses for royalty computation and settlement. ITEGA adopts and encourages the Really Simple Licensing (RSL) standard (established 2025 by a nonprofit group led by Eckart Walther) for content rights tagging. RSL is designed to establish uniform protocol for tagging ownership and usage rights for internet content, including for AI model usage tracking.

Content Attribute	Description
createdAt	ISO 8601 timestamp (YYYYMMDDHHMMSS)
expiresAt	Optional expiration timestamp set by publisher
pubMbrId	Publisher's network member ID — the royalty recipient
doi	Optional Digital Object Identifier (doi.org)
rslTag	Really Simple Licensing tag (rslstandard.org)
pageClass	Numeric value class from a system-wide table. Represents wholesale royalty price.
markupRatio	Retail markup ratio supplied by home base in real time, enabling user to see retail price before purchase.

3.4 The Wholesale-Retail Pricing Model

A distinctive and economically important feature of the Newshare Network is the wholesale-retail pricing model. Publishers set a wholesale price (pageClass) for their content. Home bases set a retail markup ratio (markupRatio) for their users. When a user requests content, the publisher's server receives the user's markupRatio in real time and can display the actual retail price — including the home base's markup — before the user commits to a purchase.

> *Example: The New York Times charges $1/month for a French-language content bundle. Le Figaro sets a wholesale price of €0.05 per article. When a NYT subscriber clicks a Le Figaro article, the network logs a €0.05 charge to NYT and a €0.05 credit to Le Figaro. Whether NYT charges its user anything beyond the $1/month bundle price is NYT's business decision. Two users from different home bases may see different retail prices for the same article simultaneously — just as the same toaster sells for different prices at different stores.*

This model mirrors the physical wholesale-retail marketplace and is directly analogous to how online advertising exchanges already operate in real time. It enables a genuine free market for digital information without any single entity controlling pricing.

4. Logging and Settlement Service

4.1 Design Principles

- The ALS knows users only by their opaque networkUserId. It never has access to names, email addresses, or financial information.
- The ALS operates as a “common shared service” analogous to DNS — logically centralized but physically distributed and hierarchical for fault tolerance and performance.
- Clickstream data is never sold or provided to third parties. It is returned only to the user’s home base (for billing and analysis) and to the content publisher (aggregated totals only, for audit purposes).
- Publishers receive aggregated usage reports sorted only by home-base ID — not by individual user. They cannot correlate individual users across the network.
- Attempts by publishers to correlate individual users through network data are prohibited by ITEGA membership rules and subject to sanction.

4.2 Log Record Format

Each value-exchange event logged to the ALS contains the following fields, derived from the original Clickshare extended Common Log Format:

Field	Description
timestamp	ISO 8601 event timestamp
networkUserId	Opaque pairwise pseudonymous user ID
homeBaseId	Network ID of user’s home base
pubMbrId	Network ID of content-serving publisher
resourceId	URL or DOI of the resource accessed
pageClass	Numeric value class (wholesale royalty amount)
serviceClass	User’s NetworkGroupId at time of event
markupRatio	Retail markup ratio applied by home base
eventType	content_access | ad_view | subscription_credit | reward
sessionId	ALS-assigned session identifier (not linkable to PII)

4.3 Settlement Process

Settlement runs on a configurable cycle (daily, weekly, or monthly; weekly is recommended for the pilot). The settlement process:

4. Aggregates all log records for the settlement period.
5. Groups records by homeBaseId to compute total charges owed by each home base to the network.
6. Groups records by pubMbrId to compute total credits owed to each publisher.

7. Computes ITEGA’s transaction fee (a small percentage of settled value, the network’s revenue model).
8. Generates ACH debit/credit files for processing through the U.S. banking Automated Clearing House network.
9. Delivers itemized usage reports to home bases (full clickstream by their users) and to publishers (aggregated totals by source home base).

The settlement service does not require real-time operation — it runs offline against the log database. This separates performance-critical authentication from batch financial processing, exactly as in the original TVS architecture.

For the pilot, modern ACH APIs (e.g., Stripe Treasury, Dwolla, or direct bank ACH integration) replace the 1996 Bank of Boston ACH interface. Stripe Connect is a particularly strong candidate given its support for marketplace payment splitting, which mirrors the home-base/publisher settlement model directly.

5. Recommended Technology Stack

5.1 Design Philosophy

The Newshare Network should be built from open-source components wherever possible, minimizing proprietary lock-in and reducing the cost of building and operating the reference implementation. The goal is a stack that a small team of developers familiar with modern web infrastructure can deploy in weeks, not months. The following recommendations reflect what is available in early 2026.

5.2 Component Recommendations

Component	Recommended Approach
OIDC / Identity Provider	Keycloak (open source, Java, Apache 2.0 license) or Authentik (open source, Python/Go). Both support OIDC, PPID, custom claims, and federation. Keycloak has stronger enterprise adoption; Authentik has a more modern admin UX. Either can serve as the home base IdSP for the pilot.
Home Base User Profile Store	PostgreSQL with a schema derived from the Clickshare Customer Profile Server schema (Section 3.2). Apache Unomi (open-source W3C Context API implementation) is a candidate for the profile management layer, given prior ITEGA exploration of this approach.
ALS — Authentication Service	Custom lightweight service built on top of the chosen OIDC provider's token validation endpoint. For the pilot, this can run as a sidecar to Keycloak/Authentik. At scale, this mirrors the original distributed CALS architecture.
ALS — Logging Service	TimescaleDB (PostgreSQL extension for time-series data) or ClickHouse for high-volume event logging. Extended Common Log Format records written via a lightweight async daemon (analogous to the original cs-logd).
ALS — Settlement Service	Python or Node.js batch process reading from the log database, generating settlement reports and ACH files. Stripe Connect API for actual payment processing in the pilot.
Publisher Integration	WordPress plugin (given the prevalence of WordPress in independent news) implementing the OIDC Relying Party flow with NetworkGroupId mapping. This is the primary publisher-side integration artifact for the Missouri pilot.
Content Rights Tagging	RSL standard metadata embedded in article HTML/JSON-LD. ITEGA to provide a tagging guide and validator.
Network Discovery	OIDC Discovery endpoint at a well-known ITEGA URL, listing certified home bases and publishers. Analogous to the original Clickshare Interchange Service (CIS) master ID database.
Demo / Consumer Experience	React-based user dashboard showing the user their session, their home base, which publishers they've visited, and their account balance. Directly implements the "user queries their clickstream" feature described in the original patent.

5.3 Deployment Architecture for Pilot

For the Missouri proof-of-concept pilot, the recommended deployment is:

- A single Keycloak or Authentik instance serving as both the ALS authentication service and a demo home base IdSP, deployed on a cloud VM (AWS, Google Cloud, or DigitalOcean).
- A PostgreSQL + TimescaleDB instance for the log and profile databases.
- A WordPress plugin deployed at 3–5 participating Missouri newspapers, implementing the OIDC RP flow and NetworkGroupId mapping.
- A Python settlement script running weekly, generating usage reports and (in the pilot) simulated ACH settlement records.
- A simple React dashboard for demo user experience.

Total infrastructure cost for the pilot is estimated at $300–$500/month in cloud hosting. Development effort is estimated at 4–6 months for a team of 2–3 developers.

6. Governance, Certification and Membership

6.1 ITEGA's Role

ITEGA is the governing authority for the Newshare Network. It does not operate services; it licenses and certifies those who do. Its role is analogous to Visa International in the credit card network, or ICANN in the domain name system: setting rules, maintaining standards, certifying participants, and enforcing compliance — without competing with the commercial parties it governs.

ITEGA's authority derives from its membership agreement, which all network participants sign as a condition of participation. The membership agreement incorporates ITEGA's Interchange Rules by reference. Violations are subject to sanction up to and including removal from the network.

6.2 Certification Tiers

ITEGA certifies three categories of participant:

Category	Description and Requirements
Content Publishers (CP)	News organizations and information services that serve content to network users. Certification requires: OIDC Relying Party implementation, NetworkGroupId support, RSL content tagging, compliance with ITEGA data handling rules, and annual audit by ITEGA-approved auditor.
Identity Service Providers (IdSP / Home Base)	Publishers or service providers that manage end-user accounts and vouch for users across the network. Higher certification bar: full OIDC Provider implementation with PPID support, profile management system, privacy policy compliance, and ACH settlement capability.
Tech Stack Providers	Companies providing software or infrastructure services to network participants. ITEGA may eventually accredit (rather than merely certify) tech stack providers who demonstrate sustained conformance.

6.3 Certification as a Near-Term Revenue Model

Importantly, ITEGA can begin generating revenue from certification before the full network is operational. Publishers who wish to demonstrate privacy-forward practices to their readers and regulators can obtain ITEGA certification independently of network participation. This is a near-term revenue path that builds ITEGA's credibility and financial sustainability while the technical network is under development.

The certification fee structure is set by the ITEGA board. Suggested initial structure: a one-time application fee, an annual certification fee scaled to publisher size, and a separate fee for the required third-party audit (conducted by ITEGA-approved auditors such as Ernst & Young or the Alliance for Audited Media).

6.4 Sanctions

The ITEGA board maintains a sanctions process for member violations. Sanctions may include warnings, fines, temporary suspension, and permanent removal from the network. The sanctions process is documented in ITEGA's Interchange Rules and is designed to ensure that the entity imposing sanctions has no competitive interest in the outcome — a key reason why the governing authority must be a nonprofit with no stake in commercial outcomes.

7. Missouri Pilot — Proof of Concept

7.1 Objective

The immediate objective is a working proof-of-concept demonstration involving a small consortium of independent Missouri newspapers, hosted at the Reynolds Journalism Institute at the University of Missouri. The pilot demonstrates the core end-to-end architecture: a user creates one account at a home base publisher, visits a second participating publisher, is recognized without re-registering, and a transaction event is logged and settled.

7.2 Proposed Participants

- Institutional host and academic partner: Donald W. Reynolds Journalism Institute (RJI), University of Missouri
- Industry convener: Missouri Press Association
- Technical operator: PubGen.AI (reference implementation, OIDC infrastructure)
- Governing authority: ITEGA
- Participating publishers: 3–5 independent Missouri newspapers (mix of small daily and weekly/digital outlets, to be recruited through MPA)

7.3 Pilot Success Criteria

10. A user can create one account at Publisher A (home base) and access content at Publisher B (outside publisher) without re-registering.
11. Publisher B receives the user's NetworkGroupId and serves content according to the user's subscription tier.
12. The ALS logs the access event with an opaque user ID — no PII transmitted to the ALS.
13. Publisher B cannot identify the user's real identity from network data.
14. Weekly settlement runs produce accurate usage reports and simulated ACH settlement records.
15. At least 50 real users complete the cross-publisher authentication flow during the pilot period.

7.4 Pilot Timeline

Phase	Activities and Milestones
Months 1–2: Setup	Deploy OIDC infrastructure (Keycloak/Authentik). Configure demo home base. Build WordPress publisher plugin v1. Recruit 3 MPA member newspapers.
Months 3–4: Integration	Deploy plugin at participating newspapers. Onboard pilot users. Stand up logging service and weekly settlement report.
Month 5–6: Pilot Operations	Live pilot with real users. Monitor, debug, iterate. Collect qualitative feedback from publishers and users.
Month 6: Evaluation	Produce pilot report. Assess against success criteria. Prepare materials for Phase 2 funding and broader rollout.

7.5 Budget Estimate

Item	Estimated Cost
Technical development (2 developers, 6 months)	$180,000 – $240,000
Project management / coordination (0.5 FTE)	$40,000 – $60,000
Cloud infrastructure (6 months)	$3,000 – $5,000
Travel, meetings, publisher outreach	$10,000 – $20,000
Legal (membership agreement, certification criteria)	$15,000 – $25,000
Contingency (10%)	$25,000 – $35,000
Total Estimated Pilot Budget	$273,000 – $385,000

8. Funding Strategy

8.1 Framing for Funders

The ask to foundations and academic partners should be framed as: fund a public-interest technical standard and reference implementation that the journalism industry can adopt without vendor lock-in. This is not a startup investment; it is infrastructure investment in a public good, analogous to funding the development of open standards for the web.

8.2 Target Funders

- Knight Foundation — has funded journalism technology infrastructure repeatedly; direct alignment with news industry focus.
- Reynolds endowment through RJI — if RJI is the institutional host, internal project funding is possible; Densmore’s prior fellowship relationship is a warm path in.
- MacArthur Foundation — funds information ecosystem and public interest technology work.
- Lenfest Institute — focused specifically on local news sustainability; strong mission alignment.
- Mozilla Foundation — funds open internet infrastructure and privacy technology.
- National Science Foundation — potential for academic research funding through RJI/Mizzou partnership.

8.3 Sustainability Model

Post-pilot, the Newshare Network is designed to be self-sustaining through:

- Transaction fees on settled value (small percentage, analogous to credit card interchange, but far lower)
- Annual certification fees from member publishers, home bases, and tech stack providers
- Audit facilitation fees
- Premium services (audience measurement data, advanced analytics) offered by licensed ALS operators

The network is explicitly not designed to generate profit for ITEGA. Revenue covers operating costs and reserves. Operators and publishers capture the commercial value.

9. Competitive Differentiation

9.1 How Newshare Differs from Existing Solutions

Solution	Key Difference from Newshare
Piano / Pico / Zuora	Commercial paywalls and subscription management tools. Single-publisher silos. No cross-publisher identity federation. No user privacy by design. Profit motive conflicts with publisher interests.
Google / Meta login ("Sign in with Google")	User identity owned by platform, not publisher. Platform learns cross-site behavior. No settlement or payment capability. Creates dependency on for-profit surveillance infrastructure.
Apple News+	Closed ecosystem. Apple captures margin. Publishers lose direct reader relationship. No cross-publisher micropayment or flexible pricing.
Solid / WebID (Berners-Lee)	Strong privacy model; user-controlled data pods. No journalism-specific settlement or payment layer. Very early ecosystem; limited publisher adoption. Complementary rather than competing.
InCommon / Internet2	Closest operational analog for higher education. ITEGA draws on this model explicitly. InCommon does not serve commercial news publishing or include a settlement/payment layer.
Newshare Network	Nonprofit governed. Privacy-by-design from architecture. Pairwise pseudonymous IDs. Wholesale-retail pricing. Per-click or subscription. Settlement via ACH. No platform intermediary. No central identity database.

10. Technical References and Prior Work

Document	Description
U.S. Patent No. 7,324,972 (expired 2022)	“Managing Transactions on a Network: Four or More Parties.” Clickshare Service Corp. Filed 2000, granted 2008. The foundational architecture document.
ITEGA Technical Design Specs v6 (Jan 2026)	23-page working specification document by Bill Densmore. The primary source document for this specification.
ITEGA SSO Network Description v4 (Jan 2026)	Detailed authentication flow specification by Richard Lerner and Bill Densmore. Source of pairwise ID scheme and NetworkGroupId design.
Clickshare Customer Profile Server Schema (2017)	JSON schema for home-base user profile storage, by Richard Lerner.
ITEGA Certification Role (Appendix A)	Description of ITEGA’s near-term certification business model.
OpenID Connect Core 1.0	https://openid.net/specs/openid-connect-core-1_0.html
W3C Verifiable Credentials Data Model 2.0	https://www.w3.org/TR/vc-data-model-2.0/
Really Simple Licensing (RSL) Standard	https://rslstandard.org/
Internet2 eduPerson Object Class Spec	https://www.internet2.edu/products-services/trust-identity/eduperson/
Apache Unomi (W3C Context API)	https://github.com/apache/unomi — candidate for home-base profile management layer.
Schema.org/Person	https://schema.org/Person — standard for person attribute vocabulary.
InCommon Federation	https://incommon.org — operational analog in higher education.

Appendix A: Glossary of Terms

Term	Definition
ACH	Automated Clearing House — the U.S. banking network for electronic debit and credit transfers. Used for periodic settlement among network participants.
ALS	Authentication, Logging and Settlement Service — the neutral infrastructure layer of the Newshare Network. Licensed by ITEGA; may be distributed.
CP	Content Provider — a publisher or information service serving content to network users.
DDA	Data/Demographic Aggregator — an optional component that holds anonymized user demographic data for advertisers, linked only to opaque user IDs.
IdSP	Identity Service Provider — the home base. The party that authenticates the user and manages their profile.
ITEGA	Information Trust Exchange Governing Association — the 501(c)3 nonprofit governing authority.
JWT	JSON Web Token — the standard format for OIDC ID tokens and access tokens.
NetworkGroupId	Bitmask value encoding a user's subscription tier and access rights in a standardized format recognized by all network members.
networkUserId	The pairwise pseudonymous identifier assigned by the home base for each user+publisher combination. Opaque to all parties except the home base.
OIDC	OpenID Connect — the primary identity federation protocol used by the Newshare Network.
pageClass	Numeric value class representing the wholesale royalty price of a content item, set by the publisher.
PII	Personally Identifiable Information — never transmitted to the ALS; held only by the home base.
PPID	Pairwise Pseudonymous Identifier — the OIDC standard mechanism for assigning different opaque user IDs to different relying parties.
RSL	Really Simple Licensing — open standard for tagging content ownership and usage rights.
TVS	Token Validation Service — the original name for the Clickshare authentication system, described in U.S. Patent No. 7,324,972.
USP	User Service Provider — synonym for home base / IdSP.

END OF DOCUMENT

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